

Math 421

Wednesday, November 19

Announcements

- Drop-In Hours
 - **Today** 12-1 pm VV 311
 - Thursday 9-11 am VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Homework 9 due Friday

Intermediate Value Theorem

$$f(a) < y < f(b)$$

Theorem. Let f be a continuous real-valued function on a closed interval $[a, b]$. If y is a value between $f(a)$ and $f(b)$, then there exists at least one $x \in (a, b)$ such that $f(x) = y$.

Showing there

Common application: exists. $x_0 \in (a, b)$ such that

$f(x_0) = 0$. Need to show that either:

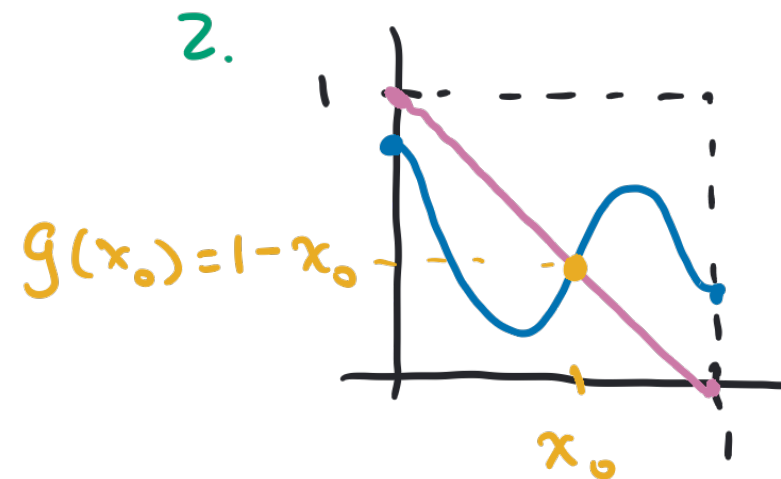
$$f(a) < 0 < f(b) \quad \text{OR} \quad f(b) < 0 < f(a).$$

Recipe for Solving IVT Problems

- I. Define a function $f(x)$.
- II. Define a number y .
- III. Choose an interval $[a, b]$. *↖ want y to be between $f(a)$ and $f(b)$.*
- IV. Establish that f is continuous on $[a, b]$. *↖ "f is a polynomial"*
"f is a combination of continuous functions"
- V. Establish that y is between $f(a)$ and $f(b)$.
↑
either
 $f(a) < y < f(b)$
or
 $f(b) < y < f(a)$
- VI. Apply the conclusion of the IVT.

Example

2.



1. Show that $x^3 + 2x - 5 = 0$ has a solution.

let $a = 0$, $b = 2$

$f(0) = -5 < 0$

$f(2) = 8 + 4 - 5 = 7 > 0$

$f(x)$ is a polynomial, so f is continuous on $[0, 2]$. Note that $f(0) = -5 < 0 < 7 = f(2)$. So, by the IVT, there exists $x_0 \in (0, 2)$

such that $f(x_0) = x_0^3 + 2x_0 - 5 = 0$.

2. Let $g: [0, 1] \rightarrow [0, 1]$ be a continuous function. Show that there

is some $x_0 \in [0, 1]$ such that $g(x_0) = 1 - x_0$.

Case 1:

if $g(0) = 1 = 1 - 0$ ✓

Case 2:

if $g(1) = 0 = 1 - 1$ ✓

common technique

Case 3:

Assume $g(0) < 1$

and $g(1) > 0$. let

$f(x) = g(x) - (1 - x)$

let $y = 0$ $f(x)$ is

a combination of continuous functions so it is continuous

on $[0, 1]$. $f(0) = g(0) - (1 - 0) = g(0) - 1 < 0$, $f(1) = g(1) - (1 - 1) = g(1) > 0$

so, $f(0) < 0 < f(1)$. So, by the IVT, there exists $x_0 \in (0, 1)$ such

that $f(x_0) = 0$. which implies $g(x_0) = 1 - x_0$.

Show there exists a solution to

- $x^2 - 1 = 0$ in $[-2, 2]$

Activity

1. Show there exists a solution to the equation $\underbrace{3x^5 - 4x^2}_{g(x)} = \underbrace{3}_y$ in the interval $[0, 2]$.

2. Prove $xe^x = 2$ for some x in $(0, 1)$. $[e > 2]$